

COMMUNICATION BETWEEN MICROSERVICES USING APACHE KAFKA

The main aim of this poc is to create two microservices and facilitate communication between them using Apache Kafka.

1. Software requirements:

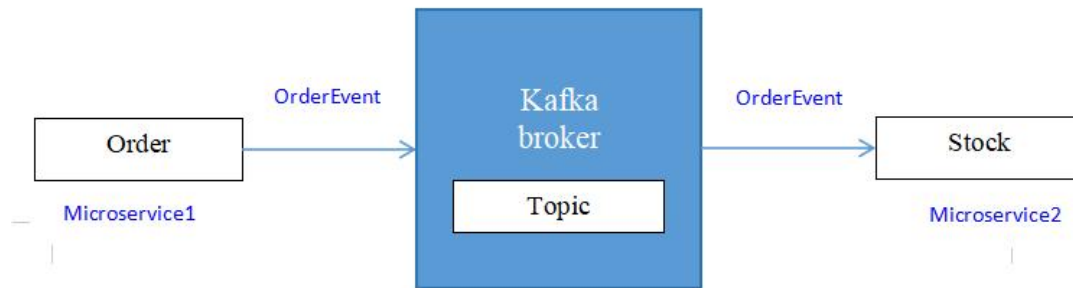
- Spring tool suite: An IDE with Eclipse-based development environment to develop Spring applications. It provides a ready-to-use environment to implement, run, deploy, and debug the application. Here we use it to create the microservices.
- Java and JDK: A programming language and development environment for building applications, applets, and components using the Java programming language. JDK includes tools useful for developing and testing programs written in the Java programming language.
- Apache Kafka: Apache Kafka is a distributed event store and stream-processing platform. It is an open-source system developed by the Apache Software Foundation written in Java and Scala. It aims to provide a unified, high-throughput, low-latency platform for handling real-time data feeds.
- Postman: A software testing API (Application Programming Interface) platform to build, test, design, modify, and document APIs. It is a simple Graphic User Interface for sending and viewing HTTP requests and responses.

2. Technologies used

- Java springboot: Java spring framework is a open source, enterprise level framework for creating standalone, production grade applications that run on java virtual machine. Java spring boot is a tool that makes developing web application and micro services with spring framework faster and easier.

3. Dataflow diagram

The below diagram shows how the message being sent by the producer microservice is being sent to the consumer microservice through the Kafka message broker.



4. Working

- As shown in the diagram we have created two microservices called order and stock. The order microservice acts as the producer or the publisher between the two services. The stock microservice acts as the consumer or the subscriber.
- First we start the zookeeper service in our command prompt with the following command,

`.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties`

- Now similarly, we also need to start the kafka server with the command

`.\bin\windows\kafka-server-start.bat .\config\server.properties`

- Now we create three microservices- Order, Stock and base domains.
- Now we can use the postman to post order details through the

1. The Base-domains service

Within this microservice we create a dto folder to pass data with multiple attributes in one shot from client to server, to avoid multiple calls to a remote server. We create all the attributes within this service that is required to send the order details. We create two classes within this, Order and OrderEvent. And we declare instance variables within them. In the Order class we have, orderId, name, qty and price as the variables. In the OrderEvent class contains, message, status and order as the variables. As shown in the diagram the OrderEvent will be published to the Kafka topic through the Order service publisher.

2. The Order service

This microservice acts as the producer/publisher as shown in the diagram. This microservice lets us send the details of the order placed into the kafka broker.

3. The Stock service

This microservice acts as the consumer/subscriber as shown in the diagram. It receives the messages sent by the order service through the kafka broker.